

Linear Algebra Primer



Gastao Cruz^a, Burhaneddin Yaman^{b,c}, Mehmet Akçakaya^{b,c}, Mariya Doneva^d, and Claudia Prieto^a

^a*School of Biomedical Engineering and Imaging Sciences, King's College London, London, United Kingdom*

^b*Department of Electrical and Computer Engineering, University of Minnesota, Minneapolis, MN, United States*

^c*Center for Magnetic Resonance Research, University of Minnesota, Minneapolis, MN, United States*

^d*Philips Research, Hamburg, Germany*

A.1 Vector spaces

Definition (Vector space). A vector space V is a set that is defined over a field F ($\mathbb{R}$ or $\mathbb{C}$) with two operations:

- vector addition: $\mathbf{v}, \mathbf{w} \in V \longrightarrow \mathbf{v} + \mathbf{w} \in V$
- scalar multiplication: $\mathbf{v} \in V, \alpha \in F \longrightarrow \alpha \mathbf{v} \in V$

that satisfy the following properties:

- commutativity of vector addition: $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
- associativity of vector addition: $\mathbf{v} + (\mathbf{w} + \mathbf{y}) = (\mathbf{v} + \mathbf{w}) + \mathbf{y}$
- additive identity: $\mathbf{v} + \mathbf{0} = \mathbf{v}$
- additive inverse: $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- associativity of scalar multiplication: $(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v})$
- distributivity of vector sums: $\alpha(\mathbf{v} + \mathbf{w}) = \alpha\mathbf{v} + \alpha\mathbf{w}$
- distributivity of scalar sums: $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$
- identity for scalar multiplication: $1\mathbf{v} = \mathbf{v}$.

A.1.1 Linear independence

Definition. A subset of vectors $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subset V$ is linearly dependent if there exists $\{\alpha_1, \dots, \alpha_n\} \in F$, not all zero, such that $\sum_{k=1}^n \alpha_k \mathbf{v}_k = \mathbf{0}$.

Definition. A subset of vectors $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subset V$ is linearly independent if it is not linearly dependent. In other words, the equation $\sum_{k=1}^n \alpha_k \mathbf{v}_k = \mathbf{0}$ can only be satisfied if $\alpha_k = 0$ for all $k \in \{1, \dots, n\}$.

A.1.2 Span

Definition. The span of $S \subset V$ is the set of all finite linear combinations in S . More formally,

$$\text{span}(S) = \left\{ \sum_{k=1}^n \lambda_k \mathbf{v}_k \mid n \in \mathbb{Z}^+, \mathbf{v}_k \in S, \lambda_k \in F \right\}.$$

If $V = \text{span}(S)$, S is called a spanning set of V .

A.1.3 Basis

Definition. A basis B of a vector space V is a linearly independent subset of vectors in V that spans V . As a result, $B \subset V$ is called a basis if every element of V can be uniquely written as a linear combination of vectors in B .

A.1.4 Normed space

Definition. A normed space is a vector space in which a norm is defined. Norm is a real-valued function that maps a vector space to real numbers and denoted with $\|\cdot\| : V \longrightarrow \mathbb{R}$ subject to:

- (i) $\|\mathbf{v}\| \geq 0$ and $\|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}$
- (ii) $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$
- (iii) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangle inequality).

Definition (ℓ_p norms). For an n -dimensional vector $\mathbf{x} \in \mathbb{C}^n$, the ℓ_p norm is defined as

$$\|\mathbf{x}\|_p = \left(\sum_{k=1}^n |x_k|^p \right)^{\frac{1}{p}}.$$

Remark. Note two special ℓ_p norms that will be used throughout the text:

- ℓ_1 norm: $\|\mathbf{u}\|_1 = \sum_{i=1}^n |u_i|$
- ℓ_2 norm: $\|\mathbf{u}\|_2 = \sqrt{\sum_{i=1}^n |u_i|^2}$.

A.1.5 Inner product space

Definition. An inner product space is a vector space in which an inner product is defined. An inner product is a map, $\langle \cdot, \cdot \rangle : V \times V \longrightarrow F$ that satisfies:

- (i) $\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle}$ (conjugate symmetry)
- (ii) $\langle \alpha\mathbf{u}, \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle$
- (iii) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
- (iv) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0 \iff \mathbf{u} = \mathbf{0}$.

Remark. It is worth noting that:

- A norm can be defined based on the inner product as $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$
- For $\mathbb{R}^n$, which is a Euclidean space, the dot product is the inner product, defined as $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i$.

A.2 Matrix theory

An $m \times n$ matrix has m rows and n columns. The $(i, j)^{th}$ entry of a matrix $\mathbf{A}$ is denoted with $\mathbf{A}_{ij}$.

Remark. A matrix can be used to represent a linear mapping/transformation between two vector spaces of finite dimension n and m . Consider a linear mapping T from $\mathbb{C}^n$ to $\mathbb{C}^m$. Then, one can find a matrix

$\mathbf{A} \in \mathbb{C}^{m \times n}$ such that

$$T(\mathbf{x}) = \mathbf{A}\mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{C}^n.$$

In order to explicitly write the $\mathbf{A}$ corresponding to the linear mapping T , the canonical basis elements $\mathbf{e}^k \in \mathbb{C}^n$ can be used. Here, $\mathbf{e}^k$ is a vector that takes the value 1 in its k^{th} coordinate and 0 elsewhere. Using the canonical basis, $\mathbf{A}$ can be explicitly written using

$$\mathbf{A} = \begin{bmatrix} T(\mathbf{e}^1) & T(\mathbf{e}^2) & \dots & T(\mathbf{e}^n) \end{bmatrix}$$

Definition (Range space). The linear span of columns of a matrix. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the range (or column) space of matrix $\mathbf{A}$ is denoted as $\text{Ran}(\mathbf{A})$ and defined as

$$\text{Ran}(\mathbf{A}) = \{\mathbf{A}\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m.$$

Definition (Null space). Null space of a matrix $\mathbf{A}$ is denoted as $\text{Null}(\mathbf{A})$ (or $N(\mathbf{A})$) and defined as

$$\text{Null}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n.$$

Definition (Rank). Number of linearly independent columns of a matrix. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, rank of the matrix $\mathbf{A}$ is denoted as $\text{rank}(\mathbf{A})$ and defined as

$$\text{rank}(\mathbf{A}) = \dim(\text{Ran}(\mathbf{A})) \leq n,$$

where $\dim(\cdot)$ denotes the dimension of the matrix.

Remark. The matrix $\mathbf{A}$ is full-rank if $\text{rank}(\mathbf{A}) = \min\{m, n\}$. Otherwise, the matrix $\mathbf{A}$ is rank-deficient.

Definition (Transpose). A transpose of a matrix $\mathbf{A}$, denoted by $\mathbf{A}^T$, switches the row and column indices of the matrix $\mathbf{A}$, i.e.,

$$(\mathbf{A}^T)_{ij} = \mathbf{A}_{ji}.$$

Definition (Conjugate-transpose). The conjugate-transpose, also known as the Hermitian transpose, of a matrix $\mathbf{A}$ is obtained by taking the transpose and then taking the complex conjugate of each entry. It is often denoted as $\mathbf{A}^H$ or $\mathbf{A}^*$.

Remark. For real matrices, the conjugate transpose is equivalent to the transpose $\mathbf{A}^H = \mathbf{A}^T$.

Definition (Inverse). Inverse of a $n \times n$ square matrix $\mathbf{A}$ is $\mathbf{A}^{-1}$ such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

Definition (Pseudo-inverse). The (Moore–Penrose) pseudoinverse, denoted $\mathbf{A}^\dagger$, generalizes the notion of inverse of a square matrix to arbitrary matrices. The pseudoinverse of a matrix $\mathbf{A}$, with linearly independent columns, is defined as

$$\mathbf{A}^\dagger = \left(\mathbf{A}^H \mathbf{A}\right)^{-1} \mathbf{A}^H,$$

which yields a left inverse, i.e., $\mathbf{A}^\dagger \mathbf{A} = \mathbf{I}$.

For a matrix $\mathbf{A}$, with linearly independent rows, it is defined as

$$\mathbf{A}^\dagger = \mathbf{A}^H (\mathbf{A}\mathbf{A}^H)^{-1},$$

yielding a right inverse, i.e., $\mathbf{A}\mathbf{A}^\dagger = \mathbf{I}$.

Definition (Eigenvalue and eigenvector). An eigenvalue and corresponding eigenvector of a square matrix $\mathbf{A}$ is a scalar λ and vector $\mathbf{v}$, respectively, such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Definition (Trace). The trace of a square matrix is the sum of all its diagonal elements, i.e.,

$$tr(\mathbf{A}) = \sum_{i=1}^n a_{ii}.$$

Remark. Traces and eigenvalues are related, in particular, the trace of matrix $\mathbf{A}$ is the sum of all its eigenvalues.

A.2.1 Types of matrices

Identity matrix: A square matrix with ones in the diagonal and zeros elsewhere is defined as

$$\mathbf{I}_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

Diagonal matrix: A square matrix in which all nondiagonal entries are zeros.

Symmetric matrix: A square matrix that is equal to its transpose.

Hermitian matrix: A complex square matrix that is equal to its conjugate-transpose, $\mathbf{A}^H = \mathbf{A}$, is a complex square matrix that is equal to its own conjugate transpose—that is, the element in the i -th row and j -th column is equal to the complex conjugate of the element in the j -th row and i -th column, for all indices i and j .

Unitary matrix: A square matrix, $\mathbf{U}$, is unitary if $\mathbf{U}\mathbf{U}^H = \mathbf{U}^H\mathbf{U} = \mathbf{I}$. For a unitary matrix $\mathbf{U}^H = \mathbf{U}^{-1}$.

A.2.2 Matrices with special structures

Toeplitz matrix: Entries are constant along diagonals, e.g.,

$$\mathbf{T} = \begin{bmatrix} t_1 & t_2 & t_3 & t_4 & t_5 \\ t_6 & t_1 & t_2 & t_3 & t_4 \\ t_7 & t_6 & t_1 & t_2 & t_3 \\ t_8 & t_7 & t_6 & t_1 & t_2 \\ t_9 & t_8 & t_7 & t_6 & t_1 \end{bmatrix}$$

Remark. The recall the linear convolution of two N -dimensional vectors (or sequences) is defined as

$$(\mathbf{x} * \mathbf{h})_n = \sum_{k=0}^{N-1} x_k h_{n-k},$$

where h_k is defined to be 0 for $k \notin \{0, \dots, N-1\}$. The linear convolution can be implemented via matrix multiplication with a Toeplitz matrix as follows:

$$\mathbf{x} * \mathbf{h} = \begin{bmatrix} h_0 & 0 & \dots & 0 & 0 \\ h_1 & h_0 & & \vdots & \vdots \\ h_2 & h_1 & \dots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ h_{N-1} & h_{N-2} & \dots & \vdots & h_0 \\ \vdots & \vdots & & \vdots & \vdots \\ \vdots & \vdots & & h_{N-1} & h_{N-2} \\ 0 & 0 & 0 & \dots & h_{N-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}.$$

Note here we have indexed the vector coordinates from 0 to $N-1$ for a simpler notation in the convolution output.

Hankel matrix: Entries are constant along antidiagonals, e.g.,

$$\mathbf{H} = \begin{bmatrix} h_1 & h_2 & h_3 & h_4 & h_5 \\ h_2 & h_3 & h_4 & h_5 & h_6 \\ h_3 & h_4 & h_5 & h_6 & h_7 \\ h_4 & h_5 & h_6 & h_7 & h_8 \\ h_5 & h_6 & h_7 & h_8 & h_9 \end{bmatrix}.$$

Circulant matrix: Entries in a row are cyclically right-shifted to form the next row, e.g.,

$$\mathbf{C} = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 & c_5 \\ c_5 & c_1 & c_2 & c_3 & c_4 \\ c_4 & c_5 & c_1 & c_2 & c_3 \\ c_3 & c_4 & c_5 & c_1 & c_2 \\ c_2 & c_3 & c_4 & c_5 & c_1 \end{bmatrix}$$

Remark. The recall the circular convolution of two N -dimensional vectors (or sequences) is defined as

$$(\mathbf{x} *_c \mathbf{h})_n = \sum_{k=0}^{N-1} x_k h_{(n-k) \bmod N}.$$

Circular convolution can be implemented via matrix multiplication with a circulant matrix as follows:

$$\mathbf{x} *_c \mathbf{h} = \begin{bmatrix} h_0 & h_{N-1} & \dots & h_2 & h_1 \\ h_1 & h_0 & h_{N-1} & \dots & h_2 \\ h_2 & h_1 & h_0 & \dots & h_3 \\ \vdots & \vdots & & \vdots & \vdots \\ h_{N-1} & h_{N-2} & \dots & \vdots & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{bmatrix}$$

Discrete Fourier transform (DFT) matrix: DFT matrix implements the DFT as a transformation matrix. For the N -point DFT, the unitary DFT matrix, $\mathbf{F}$, is the $N \times N$ matrix defined as

$$\mathbf{F} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \dots & \omega^{2(N-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \dots & \omega^{3(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \omega^{3(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{bmatrix},$$

where $\omega = e^{-\frac{i2\pi}{N}}$.

Remark. Any circulant matrix is diagonalizable by the DFT matrix. In particular, we have

$$\mathbf{C} = \begin{bmatrix} c_0 & c_{N-1} & \dots & c_2 & c_1 \\ c_1 & c_0 & c_{N-1} & \dots & c_2 \\ c_2 & c_1 & c_0 & \dots & c_3 \\ \vdots & \vdots & & \vdots & \vdots \\ c_{N-1} & c_{N-2} & \dots & \vdots & c_0 \end{bmatrix} = \mathbf{F}^H \mathbf{D}_C \mathbf{F},$$

where $\mathbf{D}_C$ is a diagonal matrix whose m^{th} diagonal entry is the m^{th} DFT coefficient of $\mathbf{c} = [c_0, \dots, c_{N-1}]^T$. In effect, this implements the convolution theorem, i.e., convolution in image domain as a multiplication in Fourier domain.

Casorati matrix: Given a 3D multidimensional array $\mathcal{X}$ of size $I \times J \times K$, Casorati matrix is formed by vectorizing the frontal slices of the $\mathcal{X}$, i.e., vectorizing $\{\mathcal{X}(:, :, k)\}_{k=1}^K$, leading to a matrix with a size of $IJ \times K$.

A.2.3 Special matrix products

Kronecker product: The Kronecker product between two matrices $\mathbf{A} \in \mathbb{R}^{I \times J}$ and $\mathbf{B} \in \mathbb{R}^{K \times L}$, denoted by $\mathbf{A} \otimes \mathbf{B}$, is defined as

$$\begin{aligned} \mathbf{A} \otimes \mathbf{B} &= \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1J}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2J}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}\mathbf{B} & a_{I2}\mathbf{B} & \cdots & a_{IJ}\mathbf{B} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{a}_1 \otimes \mathbf{b}_1 & \mathbf{a}_1 \otimes \mathbf{b}_2 & \mathbf{a}_1 \otimes \mathbf{b}_3 & \cdots & \mathbf{a}_1 \otimes \mathbf{b}_L \\ \mathbf{a}_2 \otimes \mathbf{b}_1 & \mathbf{a}_2 \otimes \mathbf{b}_2 & \mathbf{a}_2 \otimes \mathbf{b}_3 & \cdots & \mathbf{a}_2 \otimes \mathbf{b}_L \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_I \otimes \mathbf{b}_1 & \mathbf{a}_I \otimes \mathbf{b}_2 & \mathbf{a}_I \otimes \mathbf{b}_3 & \cdots & \mathbf{a}_I \otimes \mathbf{b}_L \end{bmatrix}. \end{aligned}$$

Hadamard product: The element-wise matrix product. Hadamard product between two matrices $\mathbf{A} \in \mathbb{R}^{I \times K}$ and $\mathbf{B} \in \mathbb{R}^{I \times K}$ is denoted by $\mathbf{A} \circ \mathbf{B}$ and defined as

$$\mathbf{A} \circ \mathbf{B} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} & \cdots & a_{1K}b_{1K} \\ a_{21}b_{21} & a_{22}b_{22} & \cdots & a_{2K}b_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ a_{I1}b_{I1} & a_{I2}b_{I2} & \cdots & a_{IK}b_{IK} \end{bmatrix}.$$

A.2.4 Matrix decompositions

Eigenvalue decomposition (EVD): EVD of a $n \times n$ matrix $\mathbf{A}$ is the factorization of $\mathbf{A}$ into the product of three matrices as

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1},$$

where $\mathbf{Q}$ is a $n \times n$ square matrix whose columns are eigenvectors of $\mathbf{A}$ and $\mathbf{\Lambda}$ is a $n \times n$ diagonal matrix whose elements are eigenvalues of $\mathbf{A}$.

Singular value decomposition (SVD): SVD of a $m \times n$ matrix $\mathbf{A}$ is the factorization of $\mathbf{A}$ into the product of three matrices as

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T,$$

where $\mathbf{U}$ is a $m \times m$ unitary matrix, $\mathbf{D}$ is a $m \times n$ diagonal matrix with nonnegative entries on the diagonal, and $\mathbf{V}$ is a $n \times n$ unitary matrix. SVD of the matrix $\mathbf{A}$ can be also written as

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T,$$

where $\sigma_i = \mathbf{D}_{ii}$ are the singular values of $\mathbf{A}$, $\mathbf{u}_i$ and $\mathbf{v}_i$ are the columns of $\mathbf{U}$ and $\mathbf{V}$, respectively, and $r \leq \min\{m, n\}$ is the rank of $\mathbf{A}$.

Definition (Condition number). Measures amount of change in output of a function based on a small change in the input argument. Given a matrix $\mathbf{A}$, the condition number is denoted with $\kappa(\mathbf{A})$ and defined

as

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})},$$

where $\sigma_{\max}(\mathbf{A})$ and $\sigma_{\min}(\mathbf{A})$ are maximum and minimum singular values of $\mathbf{A}$, respectively.

Remark. If a problem has a low condition number, it is *well-conditioned*. Otherwise, it is *ill-conditioned*.

Definition (Principal component analysis (PCA)). PCA is a dimensionality reduction approach that finds directions of maximum variance in high-dimensional data and projects it onto a lower-dimensional subspace. In particular, PCA is a unitary linear transformation converting the data into a new coordinate system. These new orthogonal axes or principal components represent directions of maximum variance, where the greatest variance of data lies on the first coordinate, the second greatest variance in the second coordinate, and so on.

A.2.5 Matrix norms

Definition. A matrix norm is a function $f: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ that satisfies the following properties for $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$:

- (i) $\|\mathbf{A}\| \geq 0$ and $\|\mathbf{A}\| = 0 \iff \mathbf{A} = \mathbf{0}_{m \times n}$
- (ii) $\|\alpha \mathbf{A}\| = |\alpha| \|\mathbf{A}\|$
- (iii) $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$ (triangle inequality).

Definition (p-norm). Given a matrix $\mathbf{A}$ in $\mathbb{C}^{m \times n}$, the set of matrix norms is defined as

$$\|\mathbf{A}\|_p = \max_{x \in \mathbb{C}^n, x \neq 0} \frac{\|\mathbf{A}x\|_p}{\|x\|_p}.$$

The matrix norm $\|\cdot\|_p$ is induced by the vector norm $\|\cdot\|_p$. The induced matrix norms for the special cases of $p = 1, 2, \infty$ is

$$\begin{aligned} \|\mathbf{A}\|_1 &= \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|, \\ \|\mathbf{A}\|_2 &= \sqrt{\lambda_{\max}(\mathbf{A}^H \mathbf{A})} = \sigma_{\max}(\mathbf{A}), \\ \|\mathbf{A}\|_\infty &= \max_{i=1, \dots, m} \sum_{j=1}^n |a_{ij}|, \end{aligned}$$

where $\lambda_{\max}(\cdot)$ denotes the largest eigenvalue.

Definition (Frobenius norm). Given a matrix $\mathbf{A}$, the Frobenius norm is defined as the square root of the sum of the absolute squares of all the matrix entries, i.e.,

$$\|\mathbf{A}\|_F = \left(\sum_{j=1}^n \sum_{i=1}^m |a_{ij}|^2 \right)^{1/2}$$

$$= \sqrt{\text{trace}(\mathbf{A}^H \mathbf{A})} = \sqrt{\sum_{i=1}^{\min\{m,n\}} \sigma_i(\mathbf{A})^2}.$$

Definition (Nuclear norm). The nuclear norm, also known as the trace norm or the Schatten 1-norm, is defined as

$$\|\mathbf{A}\|_* = \text{trace}(\sqrt{\mathbf{A}^H \mathbf{A}}) = \sum_{i=1}^{\min\{m,n\}} \sigma_i(\mathbf{A}).$$

A.3 Tensors

Definition. Tensors are multidimensional arrays indexed with three or more indices.

A.3.1 Tensor properties

Definition (Order). The order of a tensor is the number of dimensions. An n -order tensor is given as $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$. For instance, an entry of a third order tensor $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$ is shown with $\mathcal{X}_{i,j,k} \in \mathbb{R}$ for each $i \in \{1, \dots, I\}$, $j \in \{1, \dots, J\}$, $k \in \{1, \dots, K\}$.

Definition (Slice). Two-dimensional sections defined by fixing all except two indices. For a third-order tensor, horizontal, lateral, and frontal slices are denoted by $\mathcal{X}_{i::}$, $\mathcal{X}_{:,j:}$, and $\mathcal{X}_{::k}$, respectively.

Definition (Fiber). Fibers are defined by fixing every index but one. A third-order tensor has column, row and tube fibers, denoted by mode-1 (column) fiber: $\mathcal{X}_{:,j,k}$, mode-2 (row) fiber: $\mathcal{X}_{i,:,k}$, mode-3 (tube) fiber: $\mathcal{X}_{i,j,:}$.

Definition (Matricization). Matricization, also known as unfolding, is the process of transforming a tensor into a matrix by taking all slices along one direction and stacking them together. A third-order tensor has three unfoldings defined by the order of stacking the slices. The three unfolding modes are

$$\mathcal{X}_{(1)} \in \mathbb{R}^{I \times JK}, \mathcal{X}_{(2)} \in \mathbb{R}^{J \times IK}, \mathcal{X}_{(3)} \in \mathbb{R}^{K \times IJ}.$$

Definition (Vectorization). The process of flattening a tensor into a vector, denoted by $\text{vec}(\mathcal{X})$.

A.3.2 Tensor products

Definition (Inner product). An inner product on a tensor is defined by the entry-wise product of the two tensors and summing them up. For third-order tensors, the inner product is defined as

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{ijk} \mathcal{X}_{ijk} \mathcal{Y}_{ijk}.$$

Definition (n-Mode product). The n -mode (matrix) product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{X} \times_n \mathbf{U}$. The n -mode product is defined as

$$(\mathcal{X} \times_n \mathbf{U})_{i_1 \cdots i_{n-1} j i_{n+1} \cdots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 i_2 \cdots i_N} u_{j i_n}.$$

Definition (Outer product). Given we have three vectors $\mathbf{a} \in \mathbb{R}^I$, $\mathbf{b} \in \mathbb{R}^J$, and $\mathbf{c} \in \mathbb{R}^K$, their outer product is defined as

$$\mathcal{X} = \mathbf{a} \odot \mathbf{b} \odot \mathbf{c} \in \mathbb{R}^{I \times J \times K}, \text{ where } \mathcal{X}(i, j, k) = a_i b_j c_k,$$

where $\odot$ denotes the outer product.

A.3.3 Tensor ranks

Definition (Rank-one). A rank-one N -order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is an outer product of N vectors:

$$\mathcal{X} = \mathbf{a}^{(1)} \odot \mathbf{a}^{(2)} \odot \cdots \odot \mathbf{a}^{(N)}.$$

Definition (Rank). The rank of a tensor $\mathcal{X}$ is the minimum number of rank-one tensors that can produce the tensor $\mathcal{X}$ as their summation.

A.3.4 Tensor decompositions

CANDECOMP/PARAFAC decomposition. CP Decomposition factorizes a tensor into a sum of rank-one components. CP decomposition for a third-order tensor, $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$, is defined as

$$\mathcal{X} = \sum_{r=1}^R \mathbf{a}_r \odot \mathbf{b}_r \odot \mathbf{c}_r \iff \mathcal{X}(i, j, k) = \sum_{r=1}^R \mathbf{a}_r(i) \mathbf{b}_r(j) \mathbf{c}_r(k),$$

where R is the rank of the tensor and $\mathbf{a}_r \in \mathbb{R}^I$, $\mathbf{b}_r \in \mathbb{R}^J$, and $\mathbf{c}_r \in \mathbb{R}^K$ for $r = 1, \dots, R$. The element-wise CP decomposition can be written as

$$\begin{aligned} \mathcal{X}(i, j, k) &= \sum_{r=1}^R \mathbf{a}_r(i) \mathbf{b}_r(j) \mathbf{c}_r(k) \\ &= \sum_{r=1}^R \mathbf{A}(i, r) \mathbf{B}(j, r) \mathbf{C}(k, r), \forall i \in 1, \dots, I, j \in 1, \dots, J, k \in 1, \dots, K, \end{aligned}$$

where factor matrices $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_R]$, $\mathbf{B} := [\mathbf{b}_1, \dots, \mathbf{b}_R]$, and $\mathbf{C} := [\mathbf{c}_1, \dots, \mathbf{c}_R]$.

Tucker decomposition. Tucker decomposition is a form of higher-order singular-value decomposition method. It decomposes a tensor into a core tensor with a matrix multiplied along each mode. Tucker

decomposition for a third-order tensor, $\mathcal{X} \in \mathbb{R}^{I \times J \times K}$, is defined as

$$\mathcal{X} = \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R \mathbf{G}(p, q, r) \mathbf{U}(:, p) \odot \mathbf{V}(:, q) \odot \mathbf{W}(:, r),$$

where $\mathbf{G} \in \mathbb{R}^{P \times Q \times R}$ is a core tensor, $\mathbf{U} \in \mathbb{R}^{I \times P}$, $\mathbf{V} \in \mathbb{W}^{J \times Q}$, and $\mathbf{C} \in \mathbb{R}^{K \times R}$ are the factor matrices that are obtained through principal components along corresponding modes. The element-wise Tucker decomposition can be written as

$$\mathcal{X}(i, j, k) = \sum_{p=1}^P \sum_{q=1}^Q \sum_{r=1}^R \mathbf{G}(p, q, r) \mathbf{U}(i, p) \odot \mathbf{V}(j, q) \odot \mathbf{W}(k, r).$$